Using Exponentials in Modelling

Exam code: 9MA0

Download PDF

Written by: Paul
Updated on 25 June 2025

Did this video help you? Yes No

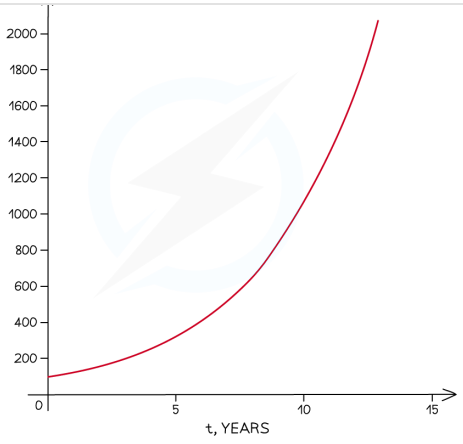
Help me understand

Using exponentials in modelling

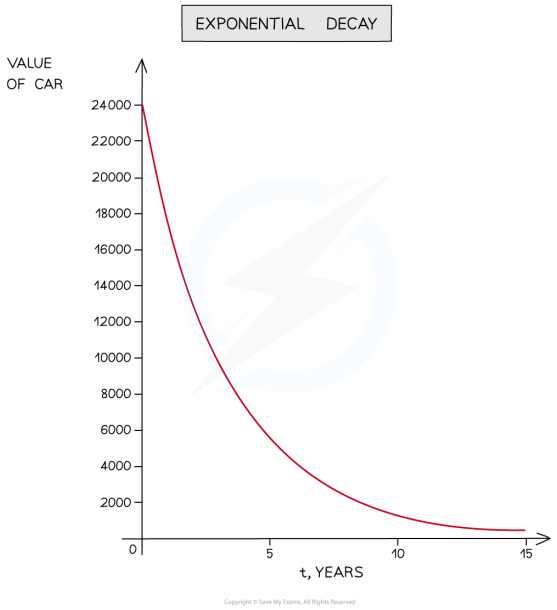
What is exponential modelling?

- Exponential modelling is where real-life situations are modelled using exponential growth and decay
- Exponential growth is typically used for ...
  - ... population (animals and humans) increase
  - ... the value of an investment under compound interest

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- Exponential decay is typically used for …
  - … levels of radioactivity
  - … quantity of a drug in a person's bloodstream
  - … depreciation (eg value of a standard car)



- It is important to consider if a model is only relevant for a certain period of time
- Can a population increase forever?
  - Will a very old car still be worth a small amount of money?

- For example: Consider a substance that initially contains 2000 radioactive atoms where the number of radioactive atoms, N, at time, t hours, is modelled by $N = 2000e^{-kt}$
  - **y** is **N**
  - **A = 2000** – the initial number of radioactive atoms
  - **k > 0** so the negative indicates this must be exponential **decay**
  - **t** is used (rather than **x**) as the situation is time-dependent

- Growth $\mathbf{y = Ae^{kt}}$
- Decay $\mathbf{y = Ae^{-kt}}$

- **A** (constant) is the starting or initial value
  - When $\mathbf{t = 0}$,
    $\mathbf{e^{kt} = e^{0} = 1}$
    so $\mathbf{y = A}$

- **k** (constant) determines the **rate** of growth/decay
- Many modelling problems are about **rates of change** –
  - If $\mathbf{y = Ae^{kt}}$
    then $\mathbf{dy/dt = Ake^{kt}}$
  - If $\mathbf{y = Ae^{-kt}}$
    then $\mathbf{dy/dt = -Ake^{-kt}}$

e.g. A SUBSTANCE INITIALLY CONTAINS 2000 RADIOACTIVE ATOMS. THE NUMBER OF RADIOACTIVE ATOMS, N, AT TIME t HOURS, IS MODELLED BY $N = 2000\,e^{-kt}$.

GIVEN THAT AFTER 2 HOURS THERE WERE 1200 RADIOACTIVE ATOMS,

a) FIND THE VALUE OF k,

b) FIND THE NUMBER OF RADIOACTIVE ATOMS AFTER SIX HOURS

Copyright © Save My Exams. All Rights Reserved

a) AT $t = 2$, $N = 1200$,

$1200 = 2000\,e^{-2k}$

$e^{-2k} = 0.6$

$-2k = \ln 0.6$

$k = -\frac{1}{2}\ln 0.6 = 0.255$ (3 sf)

Copyright © Save My Exams. All Rights Reserved

b) $N = 2000\,e^{-2k}$ $\quad k = -\frac{1}{2}\ln 0.6 = 0.255$

AT $t=6$,

$N = 2000e^{-0.255 \times 6}$

$N = 432$

USE EXACT VALUE ON CALCULATOR USING ANS OR MEMORY

Copyright © Save My Exams. All Rights Reserved

Worked Example

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources Dashboard

Join
Past Papers
Solution Banks
Sitemap

Profile
Subscription
Settings
Log out

Company
About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects
Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.